

PROJECT REPORT
ON
Fake News Detection In Deep Learning

DESH BHAGAT UNIVERSITY
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COMPANY PROFILE

ABOUT US

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- ✓ To share knowledge of information security and create awareness in the market. The solution to clients' as per the International standard practices and governance.
- ✓ To support good business practices through continual employee training and education
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- ✓ Business Process Re-Engineering
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- ✓ Mobile Application Testing
- ✓ Wireless Penetration Testing
- ✓ Network Penetration Testing
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- ✔ **Goal Oriented, ROI-Driven Focus**
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- ✔ **Talented Designers & Expert Developers**



With the EXCELLENCE TECHNOLOGY experience the incredible services such as agile software development and the problems related to outsourcing. We comprise of the team of experienced and professionals members who with their skills efficiently get the job done and innovatively help you to transform your ideas into the successful business..

- ✔ At **EXCELLENCE TECHNOLOGY**, we have competence to expand and adjust as per client specific requirements.
- ✔ **Skilled Workforce:** At EXCELLENCE TECHNOLOGY you deal with the highly professional and proficient employees.
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- ✔ **Partnership:** EXCELLENCE TECHNOLOGY considers every client a partner. From the initial stages, you are closely involved into the procedure of technical classification, development, and testing.

OUR CLIENTS





The major **RELIGIONS**
OF THE WORLD



KEY PROFESSIONALS:

In addition to a panel of eminent consultants and advisors, we have a dedicated pool of trained Developers and Trainer, investigators, working under the guidance of professional managers. **“A Ship is as good as the crew who sail her.”** Our Technical team of professionals handling, designing & delivering of projects has a strong presence in the North India & the US. Our engineers are already working on the latest technologies like **I-Phone & Android Applications, Robotics, VLSI-VHDL, Embedded System, Networking and Cloud computing.** Some of the key professionals and advisors are listed below:

Mr. Deepak Kashyap: (Branch Manager)

He is the backbone of Excellence Technology and a man with more than 10 year rich practical experience who believes in taking up new ventures and projects. Head-IT, Global Award Winner from India for AGILE and OPEN GRAPH

implementation in **Sudan Taxation Chamber**. Divisional Award and Star Award Winner in Year 2011 for his exemplary work in process improvement for IT Service Delivery Domains. MASTERS in Computer applications and Certified from PTU Certification. Holds total of 10 Years of rich experience including 5 Years in Information Security Implementation, Maintenance and Auditing and initial over 10 years experience in Project Management, Client Relationship Management and Server, Desktop and IT Service Delivery.

Ms. Sunita Thakur : (Dean Academics)

A Woman who believes that “Honour Time & Place, then you will be honored.” She has more than 4 years solid industrial experience in a software companies & is very dashing and innovative in her technical approach. MCA, Diploma in Information Technology, expertise in search engine optimization and web designing.

Mr. Gurpreet Singh : (Technical Head - Python & GIT HUB)

A man who strongly feel that “Nothing is Impossible”. A very committed team leader who has been professionally attached with Multinational companies for more than 18 years and has lead the marketing teams in all states of North India. MCA (Engineering), in computer Science PTU Approved trained and Certified. Successfully MCA master degree Training program for a leading Technology. Holds total of 5 Years of rich experience including 3 Years in Software developer, Maintenance and Auditing and initial over 4 years experience in Project Management, Client Relationship Management and IT Service Delivery.

Miss. Anjali Sharma : (Sr. Counselor)

A woman who believes that “Challenges are what make life interesting and overcoming them is what makes life meaningful.” She has more than 3 years experience in business development. B.A. – (Punjab University) Lead Consultant - Due Diligence, Strategy, Operations, Business Continuity, Risk Management, Mergers & Acquisitions.

Mr. Akshit Dogra : (Mechanical Head)

A man believes that “don’t wait for extra ordinary opportunities, seize common occasions and make them great.” He has more than 4 years experience in marketing field. B.Tech (Mech), Engineer and Value, is approved by AICTE (HPTU). He is having total 5 years of experience in 3 years of experience in Autocad, Solidworks, Catia, Pro-e, Ansys etc.

Mr. Paramveer Singh : (Mobile Application & Web Developer)

B.Tech(CSE) – IKG-PTU, Expertise in Reviewing current systems, Presenting ideas for web development soft wares, including Application softwares, Working

closely with analysts, Senior Developers, Programmers, designers and staff, Producing detailed specifications and writing the program codes, Testing the product in controlled, real situations before going live, Preparation of training manuals for users and Maintaining the systems once they are up and running

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Name of the Student

Sharnjot Kaur

Project Title:

Fake

News

Detection

Using

Deep

Learning

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Introduction:--

Our end goal is to develop an intelligent system that accurately detects and prevents fake news using advanced machine learning and deep learning techniques.

Several misinformation spread on society through social media and lead to change the opinion of people. The detection of this distinguishes from fact in it and is very important task for society. Researchers of several areas are investigating the mounting production as well as diffusion of misinformation that rapidly infects the society. "The news that is purposely and certainly false and is able to mislead the readers" is known as fake news

[1]. According to Merriam Webster Online Dictionary, fake news is "the news reports that are intentionally false or misleading" [2]. Generally, we can say that fake news is unauthenticated information that circulates or propagated on any platform for misleading to human brain, and it includes rumors, propaganda, and satire [3] The mounting curiosity to detect the fake news fascinated to researchers due to the circulation of fake information through social media platforms. Such misinforming content is quickly spreads on social media and gains popularity. As people received the content from social media, they easily can believe on it and interpret their minds for it as reliable source of information to get trust. Due to blind trust and easily acceptance

of people for fake news information, several solemn and unenthusiastic impacted fake articles get viral and observe on society and people leads to a disparity of news ecosystem [4]. Nowadays, social media (i.e., Facebook, WhatsApp, and Twitter) have covered most of the developing countries worldwide to become the major resource for news. It is essential in economics, social developments, and politics for motivated human brain and influences the process of these in negative impact and finally targets to damage to public figures and agencies [5]. Several active researchers are enduring to detect the fake news from social media. The detection process is estimating the misleading information of any news/topic whether it is planned or accidentally affected to society. In most of the cases, the detection process deploys the machine learning algorithm to filter the news, whether it is fake or not [5]. The detection of such things is very difficult task due to slight differences between fake and real news [6]. Current scenario of messaging shows that the fake news has become an ordinary thing for it. No one can refute the content easily as it may fake; still their meaning depends on view of person [7]. The identification of fake news is an imperative subject for both public and society. The amplifying social media interaction has led the increment in number of people to unfair [8]. Problem From past few decades, social media cover globe as they become the major resource of news information, due to the accessibility, minimum cost. Other perspective illustrates the danger to expose “fake news” aimed to mislead and manipulate the readers. The extensive spread of such misinformation deploys negative impacts on people and society and becomes recently a global problem. One major issue rises in U.S 2016 presidential elections due to huge spread of fake news. Therefore, the detection of it

***** Our end goal is to develop an intelligent system that accurately detects and prevents fake news using advanced machine learning and deep learning techniques***.**

In the digital era, information dissemination has become faster and more accessible than ever before. Social media platforms, blogs, and online news portals have revolutionized how people consume news and information. However, this convenience has also led to the proliferation of misinformation, disinformation, and fake news. The spread of fake news can have serious consequences, including misleading the public, inciting violence, influencing elections, and eroding trust in legitimate media sources. Detecting fake news manually is a challenging task due to the volume and velocity of online content. Hence, there is a growing need for automated systems that can accurately detect and flag fake news in real time.

This project aims to address the fake news epidemic by leveraging machine learning and natural language processing (NLP) techniques. By analyzing textual data from news articles, our goal is to develop a model that can effectively classify news as either real or fake. This report provides a comprehensive overview of the methodology, implementation, and evaluation of the fake news detection system. The importance of this research extends to areas such as journalism, education, law, and public safety, making it a significant area of concern for governments, enterprises, and individuals alike.

on social platform transforms into an emerging research that is exciting enormous concern

Simple way:-

1. Fake news spreads quickly and can shape people's opinions in misleading ways.
2. It often comes from sources like social media, blogs, and unreliable news sites.
3. Using data analysis and AI, we can identify patterns to detect and stop fake news more effectively.

DATASET:-

- Dataset : Fake News Dataset
- Size : 21,000 news articles
- Features: Title, Text, Subject, Date
- Target : Label (FAKE/REAL)

Which provides fact-checked news articles. We collected more than 21,000 news articles to train and test our fake news detection model.

WHAT IS DEEP LEARNING :-



Machine Learning (ML) and Deep Learning are subsets of Artificial Intelligence. Deep Learning represents the next evolution in Machine Learning. In Deep Learning, the model learns through an artificial neural network that is very much similar to a human brain and this allows the model to analyse data in a structure much similar to humans do. Deep Learning models don't require a human programmer to

intervene and tell what to do with the data. It is self-capable of learning from the extraordinary amount of data provided to it.

Deep learning is a machine learning technique that teaches computers to do what comes naturally to humans: learn by example. Deep learning is a key technology behind driverless cars, enabling them to recognize a stop sign, or to distinguish a pedestrian from a lamppost. It is the key to voice control in consumer devices like phones, tablets, TVs, and hands-free speakers. Deep

learning is getting lots of attention lately and for good reason. It's achieving results that were not possible before.

In deep learning, a computer model learns to perform classification tasks directly from images, text, or sound. Deep learning models can achieve state-of-the-art accuracy, sometimes exceeding human-level performance. Models are trained by using a large set of labelled data and neural network architectures that contain many layers.

In a word, accuracy. Deep learning achieves recognition accuracy at higher levels than ever before. This helps consumer electronics meet user expectations, and it is crucial for safety-critical applications like driverless cars. Recent advances in deep learning have improved to the point where deep learning outperforms humans in some tasks like classifying objects in images.

While deep learning was first theorized in the 1980s, there are two main reasons it has only recently become useful:

Deep learning requires large amounts of labelled data. For example, driverless car development requires millions of images and thousands of hours of video.

Deep learning requires substantial computing power. High-performance GPUs have a parallel architecture that is efficient for deep learning. When combined with clusters or cloud computing, this enables development teams to reduce training time for a deep learning network from weeks to hours or less.

2. Objective

The primary objective of this project is to build an intelligent system that can:

- Automatically classify news articles as fake or real based on textual content.
- Analyze large volumes of news data efficiently.
- Achieve high accuracy, precision, and recall using advanced machine learning and NLP techniques.
- Provide insights into the linguistic and semantic patterns commonly found in fake news.
- Serve as a foundational framework for future enhancements, including multimedia analysis and real-time detection tools.
- Promote media literacy and contribute to public awareness on issues of misinformation.

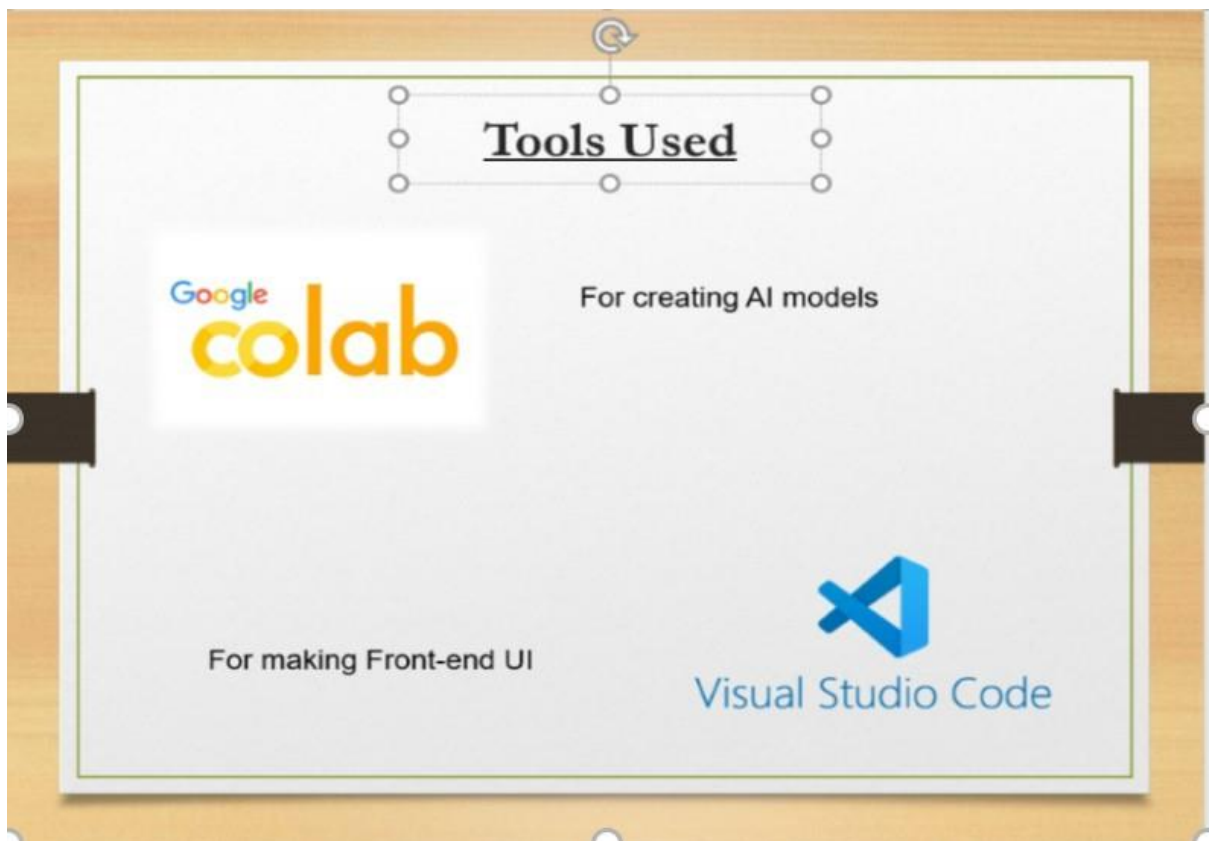
3. Problem Statement

Fake news detection presents several challenges. First, fake news articles are often crafted to mimic real news, making them difficult to detect through superficial features. Second, the

language used in fake news can be emotionally charged, misleading, or biased, requiring deep linguistic analysis. Third, the rapid spread of fake news online necessitates real-time detection and response.

Given these challenges, our problem statement is: "How can we design a scalable, accurate, and efficient machine learning model to detect fake news using textual data from news articles?" In addressing this question, we also seek to understand the linguistic structures and writing patterns that typify fake content.

4. Tools used



1. google colab:-



Google Colab (short for Colaboratory) is a free online platform provided by Google that allows users to write and run Python code through the browser. It's based on Jupyter Notebooks and is especially popular for machine learning, data analysis, and educational purposes.

Key Features:

- Free Access to GPUs and TPUs: Great for training machine learning models.
- No Setup Required: Just sign in with a Google account and start coding.
- Notebook Environment: Supports code, text, images, charts, and more in a single document.
- Cloud-Based: Automatically saves to Google Drive.

2. Visual Studio Code:-

Visual Studio Code is a free, open-source, and lightweight code editor developed by Microsoft. It supports many programming languages and is widely used by developers around the world

***Visual Studio Code (VS Code) is a popular and powerful code editor used for building projects like fake news detection. It helps you write, organize, and test your code efficiently. With features like a built-in terminal, debugging tools, and support for many programming languages, it makes development easier and faster. You can also install extensions to add extra tools, use Git for version control, and run your project offline. Overall, VS Code is a great choice for creating well-structured and professional projects.

Key Features:

- Intelligent Code Completion: Thanks to IntelliSense, which suggests code completions and shows parameter info.
- Built-in Terminal: Run command-line tools directly inside the editor.
- Extensions: Thousands of extensions for languages, themes, debuggers, linters, and more.
- Cross-Platform: Works on Windows, macOS, and Linux.

Supported Languages:

- JavaScript, Python, C++, Java, HTML, CSS, PHP, and many more.

* for version control, and run your project offline. Overall, VS Code is a great choice for creating well-structured and professional projects.


```
import re
from bs4 import BeautifulSoup as bs
import pandas
import numpy as np
import requests
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
import seaborn
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import confusion_matrix
from sklearn.linear_model import PassiveAggressiveClassifier
from sklearn.metrics import accuracy_score
from nltk.stem import WordNetLemmatizer
import wordcloud
from wordcloud import WordCloud
import matplotlib.pyplot as plt
```

Libraries Used:

1) Numpy:



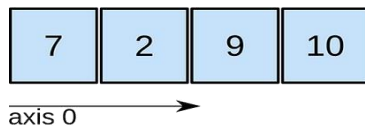
NumPy is a popular Python library used to work with numbers and data. It helps you store and handle large sets of numbers using special data structures called arrays. With NumPy, you can do math on whole arrays quickly and easily, without writing long loops. It is faster than regular Python code when working with large amounts of data.

NumPy is very useful for people working with data, like scientists, engineers, and programmers. It works well with other Python tools and is often used in projects that involve data analysis, machine learning, or scientific research. NumPy helps make your code shorter, faster, and easier to understand.

We used NumPy in our project because:

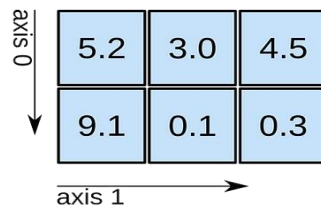
- It helps perform fast mathematical operations on data.
- It works with arrays and matrices, which are important for machine learning.
- It uses less memory and is faster than regular Python lists.
- It supports many functions for data analysis and manipulation.

1D array



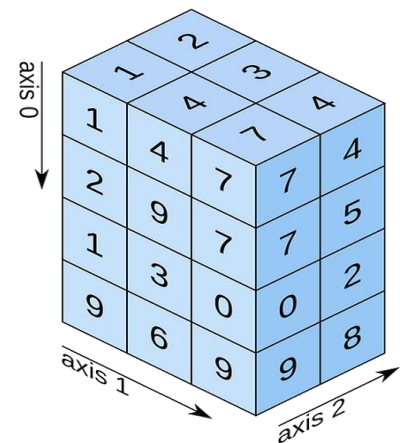
shape: (4,)

2D array



shape: (2, 3)

3D array



shape: (4, 3, 2)

Features of NumPy:

- Works with large groups of numbers using arrays
- Makes math operations fast and simple
- Lets you do math on arrays of different sizes (broadcasting)
- Saves time by not needing loops (vectorization)
- Includes functions for math, statistics, and algebra
- Can create random numbers easily
- Can load and save data from files
- Works with other Python tools like Pandas and TensorFlow
- Supports signal processing and Fourier transforms
- Can connect with faster C or C++ code if needed

2. Pandas:



Pandas is a powerful Python library used to work with structured data. It makes it easy to load, clean, analyze, and manipulate data, especially in table format like you see in Excel or CSV files. With Pandas, you can sort, filter, group, and change data quickly with just a few lines of code.

Pandas is widely used in data analysis, machine learning, and data science projects. It helps people handle real-world data easily, even if it's messy or has missing values. It also works well with other Python libraries like NumPy and Matplotlib, making it a key tool for data work in Python.

- Pandas is a library for data manipulation and analysis in Python.

- Here's why Pandas is indispensable in these tasks:

- Data Structures

- Data Cleaning

- Data Exploration

- Data Analysis

Data manipulation

In the fake news detection project, pandas was used to read and clean our dataset, making it ready for analysis and model training. It made our data handling more efficient and streamlined.

Data manipulation

In the fake news detection project, pandas was used to read and clean our dataset, making it ready for analysis and model training. It made our data handling more efficient and streamlined.

Main Features of Pandas (Simple)

- Easy to read and write data from CSV, Excel, SQL, etc.
- Handles missing data smoothly
- Supports data filtering, sorting, and grouping
- Can merge or join different datasets
- Allows reshaping and pivoting of data
- Has built-in functions for data statistics and analysis
- Works well with time series data

- Integrates with NumPy, Matplotlib, and other Python tools
- Fast performance for large datasets
- Supports custom functions and operations on data

Data Structures in Pandas :

1. Series

- One-dimensional labeled array (like a column in a table)
- Can hold data like numbers, strings, or dates
- Each item has a label (called an index)

Series

	peppers
0	3
1	2
2	0
3	1

+

Series

	carrots
0	0
1	3
2	7
3	2

=

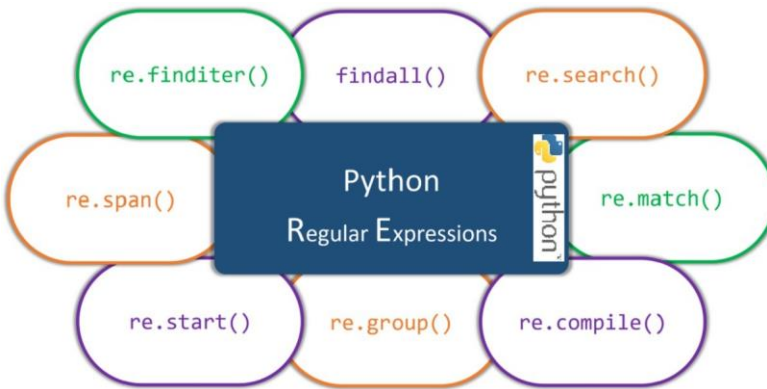
DataFrame

	peppers	carrots
0	3	0
1	2	3
2	0	7
3	1	2

2. DataFrame

- Two-dimensional labeled table (like an Excel sheet)
- Made up of rows and columns
- Each column is a Series
- Easy to view, filter, and modify rows/columns

3. Regular Expression (RegEx) :



Regular Expressions, or RegEx, are special patterns used to find or match specific parts of text. You can use RegEx to search for words, letters, numbers, or symbols in a string. It's like a smart search tool that helps you find exactly what you're looking for, even if it's not written the same way every time.

RegEx is used in many programming tasks like searching, replacing text, checking if input is valid (like an email or phone number), and splitting strings. It works in Python and many other languages like JavaScript, Java, and C++. Although it may look confusing at first, learning RegEx can save you a lot of time when working with text data.

Key Features of RegEx

- Search for patterns in text
- Replace or remove parts of a string
- Validate input (e.g., emails, passwords, phone numbers)
- Split strings using patterns
- Works across many programming languages
- Useful in data cleaning and web scraping

4.) BeautifulSoup:

BeautifulSoup

BeautifulSoup is a Python library used for web scraping, which means extracting data from websites. It helps you read and search through HTML or XML files easily. If you've ever seen a web page's source code

and wanted to grab specific information (like headlines, links, or prices), BeautifulSoup is the tool for that.

It works by taking messy or complex HTML and turning it into a structured format you can navigate using simple Python commands. It is often used along with the `requests` library to first fetch a web page, and then BeautifulSoup helps parse the content.

We used BeautifulSoup in our project because:

- It helps us scrape news articles and data from websites.
- It makes it easy to find and extract specific content, like headlines or article text.
- It works well with other libraries like requests to download web pages and parse the content.
- It's simple to use, even for beginners, and allows us to extract data quickly.

In this project, BeautifulSoup was helpful for gathering news data from websites to build our dataset. It made collecting and cleaning online content for analysis much easier.

Key Features of BeautifulSoup

- Parses HTML and XML documents
- Makes it easy to search and extract data from tags
- Handles messy or broken HTML (fault-tolerant)
- Allows searching using tag names, CSS selectors, and attributes
- Supports navigation using `.find()`, `.find_all()`, `.select()` etc.
- Works well with `requests` for downloading pages
- Converts HTML into a tree-like structure that's easy to explore

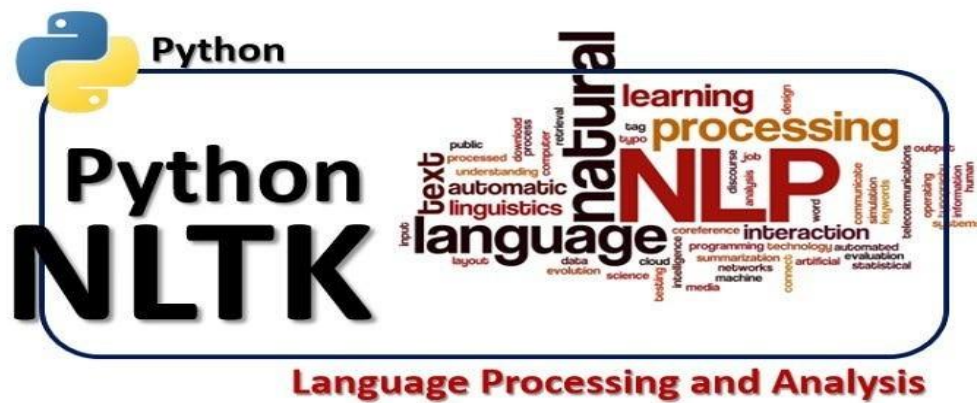
5. NLTK in NLP:



Natural Language Analysis with Python NLTK

NLTK, or Natural Language Toolkit, is a popular Python library used for working with human language data. It helps computers understand, analyze, and process text like humans do. NLTK is often used in Natural Language Processing (NLP) tasks such as breaking sentences into words, finding the root of words, or identifying parts of speech (like nouns and verbs).

NLTK is widely used in both learning and research. It comes with a collection of text data (called corpora) and built-in tools for tasks like tokenization, stemming, lemmatization, POS tagging, and text classification. It's a great library for beginners who want to explore NLP and understand how machines can work with language.



Key Features of NLTK

- Large collection of text data (corpora) and examples
- Tools for tokenizing sentences and words
- Stemming and lemmatization to reduce words to their root forms
- Part-of-speech (POS) tagging to label words as nouns, verbs, etc.
- Named entity recognition (NER) to find names of people, places, and more
- Text classification (e.g., spam detection, sentiment analysis)
- Parsing and syntax analysis
- Easily integrates with other NLP tools and libraries
- Helpful for both research and teaching NLP concepts

Used NLTK Libraries/Modules in the project:

`from nltk.tokenize import word_tokenize`

- What it does: Splits a sentence or paragraph into individual words (called *tokens*).
- Why it's useful: It's the first step in most NLP tasks – breaking text into manageable pieces.

`from nltk.corpus import stopwords`

- What it does: Provides a list of common words (like "the", "is", "and") that are usually removed from text.
- Why it's useful: These words don't carry much meaning and are often filtered out to focus on more important words.

6. Scikit-learn (sklearn)



Scikit-learn (also written as `sklearn`) is a popular Python library used for machine learning and data analysis. It provides simple and efficient tools to build machine learning models like classification, regression, clustering, and more. It is built on top of NumPy, SciPy, and Matplotlib, which makes it powerful and easy to use.

Scikit-learn is great for beginners and professionals because it has a consistent and simple API. It includes many ready-to-use algorithms and tools for model training, testing, and evaluation. You can use it to solve problems like spam detection, predicting prices, or grouping similar data.

Key Features of Scikit-learn:

- Supervised learning: Classification (e.g., SVM, KNN) and Regression (e.g., Linear Regression)
- Unsupervised learning: Clustering (e.g., K-Means), dimensionality reduction (e.g., PCA)
- Model evaluation: Accuracy, precision, recall, confusion matrix, etc.
- Model selection: Cross-validation, hyperparameter tuning with GridSearch
- Preprocessing tools: StandardScaler, MinMaxScaler, LabelEncoder, etc.
- Pipelines: Combine multiple steps (preprocessing + model) into one workflow

Main Modules in Scikit-learn :-

Module	Use
<code>sklearn.datasets</code>	Load sample datasets (like Iris, digits, etc.)
<code>sklearn.model_selection</code>	Split data, cross-validation, train-test split
<code>sklearn.preprocessing</code>	Scale and transform data
<code>sklearn.linear_model</code>	Linear models like Linear Regression, Logistic Regression
<code>sklearn.tree</code>	Decision Trees

Module	Use
<code>sklearn.ensemble</code>	Random Forest, Gradient Boosting
<code>sklearn.svm</code>	Support Vector Machines
<code>sklearn.metrics</code>	Evaluation metrics like accuracy, F1 score

7. Base64:

Base64 Encoding Table

Value	Char	Value	Char	Value	Char	Value	Char
0	A	16	Q	32	g	48	w
1	B	17	R	33	h	49	x
2	C	18	S	34	i	50	y
3	D	19	T	35	j	51	z
4	E	20	U	36	k	52	0
5	F	21	V	37	l	53	1
6	G	22	W	38	m	54	2
7	H	23	X	39	n	55	3
8	I	24	Y	40	o	56	4
9	J	25	Z	41	p	57	5
10	K	26	a	42	q	58	6
11	L	27	b	43	r	59	7
12	M	28	c	44	s	60	8
13	N	29	d	45	t	61	9
14	O	30	e	46	u	62	+
15	P	31	f	47	v	63	/

Base64 is a binary-to-text encoding scheme. It's used to represent binary data (like images or files) in a string format using only ASCII characters. This is especially useful when you need to:

- Embed images or files in HTML, CSS, or JSON.
- Transmit data in text-only environments (e.g. email, URLs).
- Store binary data in databases as text.

Variants and URL-Safe Encoding

- Standard Base64 uses + and /, which can be problematic in URLs.
- URL-safe Base64 replaces:
 - + → -
 - / → _
 - Padding = is often removed
- Many libraries support this directly:
 - Python: `base64.urlsafe_b64encode()`
 - Java: `Base64.getUrlEncoder()`

Padding Control

- Standard Base64 uses one or two = symbols for padding.

- Some systems (e.g., JWT, certain APIs) omit padding entirely.
- Libraries often allow enabling/disabling padding explicitly.

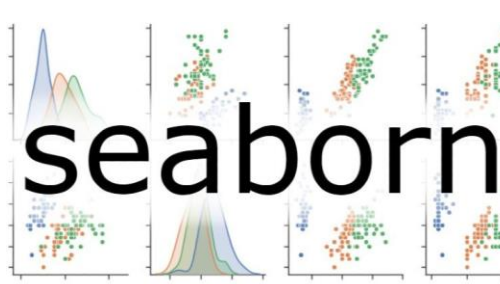
Streaming Support

- Some libraries (especially in Java, .NET, Go) provide streaming encoders/decoders for handling large files or data streams without loading everything into memory.
- Ideal for large-scale data processing (e.g., images, videos).

Line Wrapping

- MIME/Base64 (used in email) often wraps lines at 76 characters.
- Libraries support this for compatibility:
 - Java: `Base64.getMimeEncoder(76, "\r\n".getBytes())`
 - Python: `base64.encodebytes()` adds line breaks

8. Seaborn:

The image shows the word "seaborn" in a large, bold, black font. To the left of the text are four small, square plots arranged in a 2x2 grid. The top-left plot shows a violin plot with two overlapping distributions, one in blue and one in orange. The top-right plot shows a scatter plot with data points colored by a gradient from blue to red. The bottom-left plot shows a violin plot with three overlapping distributions in blue, orange, and green. The bottom-right plot shows a scatter plot with data points colored by a gradient from blue to red. The word "seaborn" is positioned over the middle of these plots.

Seaborn is a high-level data visualization library in Python built on top of Matplotlib. It provides a simple yet powerful interface for creating informative and attractive statistical graphics. Designed to work seamlessly with pandas DataFrames, Seaborn makes it easy to visualize data relationships and patterns with minimal code. It automates many aspects of plot creation, such as aggregation, color encoding, and statistical estimation, which helps in producing meaningful plots quickly and efficiently.

The library includes a wide range of plotting functions like scatter plots, line plots, box plots, violin plots, heatmaps, and more. Seaborn also supports advanced features like faceting, themes, and built-in color palettes, allowing users to create complex visualizations for multivariate data exploration. Its integration with Matplotlib means that any Seaborn plot can be further customized using Matplotlib's tools, making it a flexible choice for both quick insights and polished reports.

Key Features of Seaborn:

1. High-Level Interface

- Simplifies complex plots with minimal code.
- Intuitive syntax for visualizing relationships between variables.

2. Works Seamlessly with Pandas

- Accepts pandas DataFrames directly.
- Uses column names to reference data variables.

3. Built-in Statistical Functions

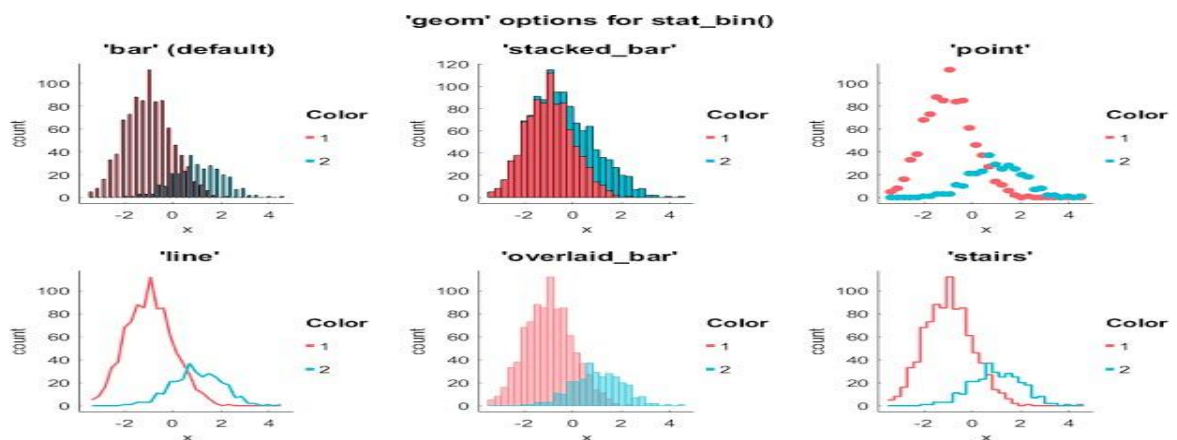
- Automatically computes:
 - Means
 - Confidence intervals
 - Linear regressions (via `regplot`, `lmplo`)
- Supports KDE (kernel density estimation), histograms, and more.

4. Attractive Default Themes

- Professional-looking plots with minimal styling needed.
- Predefined styles like `darkgrid`, `whitegrid`, etc.

5. Rich Set of Plot Types

- Supports many chart types:
 - **Distribution plots** (`histplot`, `kdeplot`, `displot`)
 - **Categorical plots** (`boxplot`, `violinplot`, `swarmplot`)
 - **Relational plots** (`scatterplot`, `lineplot`, `relplot`)



- **Matrix plots** (`heatmap`, `clustermap`)
- **Multivariate plots** (`pairplot`, `jointplot`, `lmplo`)

6. Automatic Plot Grouping

- Easily visualize grouped data with `hue`, `col`, and `row` parameters.
- Enables split-by-category comparisons without extra coding.

7. Faceting for Subplots

- Use `FacetGrid` and `catplot` to create multi-plot grids based on categorical variables.

8. Built-in Color Palettes

- Dozens of professional color palettes (`deep`, `pastel`, `muted`, `dark`, etc.)
- You can customize or create your own.

9. Full Matplotlib Compatibility

- You can extend or fine-tune any Seaborn plot using Matplotlib functions (e.g., `plt.title()`).

10. Data Visualization for Statistical Exploration

- Perfect for EDA (Exploratory Data Analysis) and data storytelling.

9. plotly:



Plotly is a powerful and interactive data visualization library in Python that is widely used in deep learning workflows for building dynamic dashboards, tracking training metrics, and visualizing model behavior. Unlike static libraries like Matplotlib, Plotly enables highly interactive plots that support zooming, panning,

and hovering, making it ideal for exploring high-dimensional data and understanding complex neural network outputs. It integrates seamlessly with Jupyter Notebooks, allowing deep learning practitioners to visually monitor real-time training progress, loss curves, and performance metrics using tools like Plotly Express or Plotly Graph Objects.

In deep learning applications, Plotly is often used alongside frameworks like TensorFlow or PyTorch to visualize training histories, activation maps, confusion matrices, and predictions. Its ability to generate 3D plots and interactive heatmaps is particularly useful when analyzing model performance across multiple parameters or classes. Plotly also supports integration with Dash (a framework by Plotly), which allows developers to create complete web-based applications for model monitoring or experimentation tracking—an essential feature for production-grade machine learning systems.

Key Features of Plotly (Python) for Deep Learning :

1. Interactive Visualizations

- Zoom, pan, hover tooltips, and selection tools built into every plot.
- Ideal for exploring model outputs and high-dimensional data.

2. Real-Time Monitoring

- Dynamically update plots during training (e.g., loss, accuracy) using Plotly in Jupyter or Dash apps.

3. 3D Plotting Support

- Supports 3D scatter, surface, mesh, and volume plots.
- Useful for visualizing embeddings (e.g., t-SNE, PCA) or volumetric data in deep learning.

4. Support for Complex Visual Types

- Includes advanced charts like:
 - Heatmaps
 - Violin plots
 - Contour plots
 - Parallel coordinates
 - Treemaps and sunbursts

5. Dash Integration

- Build complete, interactive web apps to monitor model training or performance.
- Dash + Plotly is often used for ML dashboards and MLOps tools.

6. High Customizability

- Every element of the chart (axes, labels, colors, markers, etc.) is fully customizable via `plotly.graph_objects`.

7. Plotly Express

- A high-level wrapper for quick plots with minimal code.
- Works well with Pandas DataFrames and is great for rapid experimentation.

streamlit :-



Streamlit is an open-source Python library designed to simplify the creation of interactive web applications, especially for data science, machine learning, and data visualization. It allows developers to convert Python scripts into fully functional web apps with just a few lines of code. Streamlit eliminates the need for HTML, CSS, or JavaScript knowledge, focusing solely on Python. The library provides a variety of built-in widgets, such as sliders, buttons, and input fields, making it easy to add interactivity to applications.

Additionally, it supports rendering complex visualizations, including charts, maps, and tables, directly from popular libraries like Matplotlib, Plotly, and Altair.

The simplicity of Streamlit lies in its ability to create fully interactive applications without the overhead of web development frameworks. Developers can focus on building the functionality of their apps using Python, and Streamlit handles the rest, including user interface and deployment. To run a Streamlit app, you simply write your Python code, save it as a `.py` file, and run it with the command `streamlit run app.py`. The app automatically opens in a browser, allowing for real-time interaction. This makes Streamlit an ideal tool for quickly prototyping data-driven applications and sharing them with others. Streamlit also offers a seamless experience for collaboration and sharing, as apps can be easily deployed to the web using Streamlit Cloud or hosted on platforms like Heroku or AWS. This makes it a popular choice for teams and individuals who want to showcase their data analyses, machine learning models, or prototypes in an interactive and user-friendly format. Its ability to handle dynamic inputs and update outputs in real-time without needing page refreshes enhances the user experience. Additionally, Streamlit's community is active, offering a wealth of tutorials, examples, and plugins that help developers enhance their applications and stay up-to-date with the latest features.

Fake News Detection System

Enter Any News Headline:

COVID does not exist

['FAKE']

Streamlit is a Python library that allows you to quickly create and deploy interactive web applications. It's commonly used for building data science and machine learning dashboards, but can be applied to other use cases as well. With Streamlit, you can turn a Python script into a web app with minimal effort. Here's a simple overview:

Key Features:

- **Rapid Prototyping:** You can create web apps in just a few lines of Python code.
- **Interactivity:** It supports widgets like sliders, buttons, and text boxes, making it easy to create interactive applications.
- **Data Display:** You can display various data visualizations like charts, tables, and maps.

- **Simplicity:** No need for HTML, CSS, or JavaScript; just Python.

Code and working:

1.Import packages:

It allows us to create various types of plots like line graphs, bar charts, histograms, and more.

- It provides clear, customizable visualizations to help us understand and analyze data.
- It integrates well with other libraries like NumPy and pandas for easy plotting of data stored in arrays or DataFrames.

Literature Review :--

The field of fake news detection has evolved significantly over the past decade. Early efforts focused on manual fact-checking and rule-based systems, which lacked scalability. More recent research has leveraged machine learning and deep learning to automate the detection process.

Key Approaches:

- **Statistical Models:** Naïve Bayes, Logistic Regression, and Support Vector Machines have been widely used for binary text classification. These models rely on features like term frequency, n-grams, and syntactic patterns.
- **Deep Learning Models:** Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRU) are effective in capturing the sequential nature of text. These models can understand context and dependencies between words.
- **Transformer-Based Models:** BERT (Bidirectional Encoder Representations from Transformers) has set new benchmarks in NLP tasks. By pre-training on large corpora, BERT captures bidirectional context and semantics, making it highly effective for fake news detection.
-
- Behavior analysis, and content-based filtering has shown promise in improving accuracy.
- Several datasets have also been introduced to facilitate research in this area, including LIAR, FakeNewsNet, and Kaggle Fake News Challenge datasets. These datasets vary in size, structure, and labeling techniques, providing a broad platform for experimentation.
- **Data Collection**
- We used the Kaggle Fake News dataset, which contains 21,000+ labeled news articles. Each record includes fields such as:

- Title: The headline of the news article.
- Text: The main content of the article.
- Subject: The fake news detection (e.g., politics, world).
- Label: 1 for fake news and 0 for real news.
- Other potential datasets such as the LIAR dataset provide additional granularity, including metadata about the source, author, and context, which can enhance model robustness.

Data Collection :-

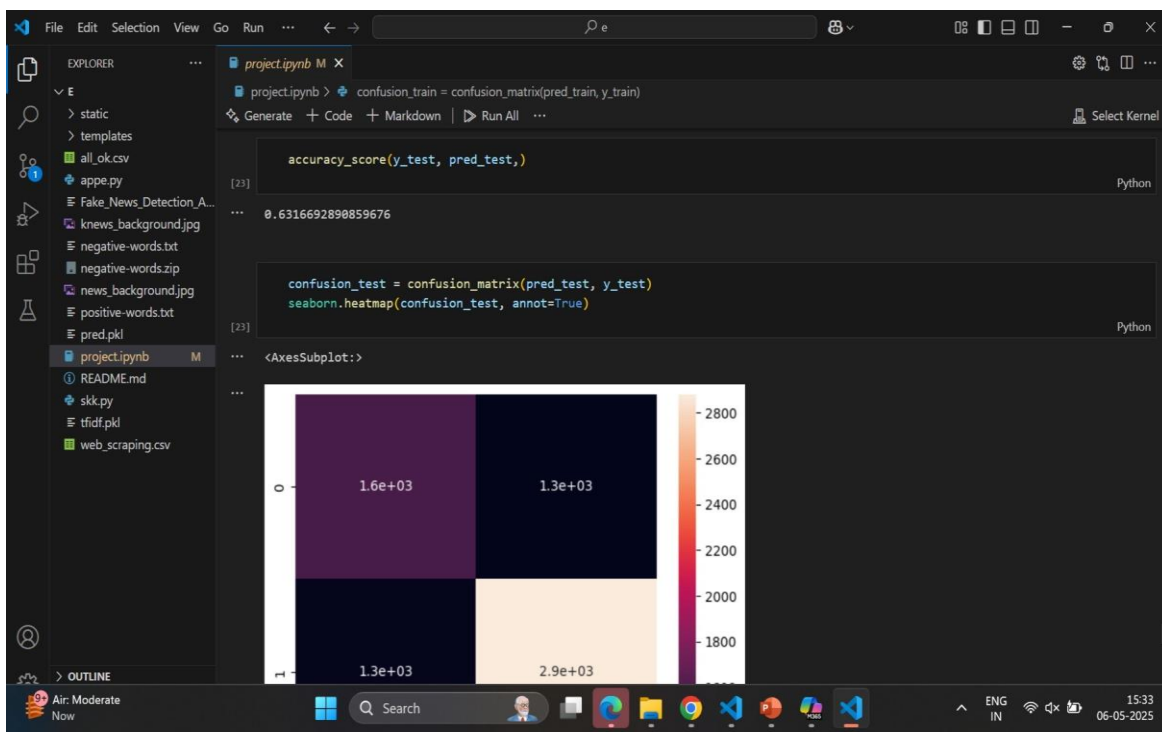
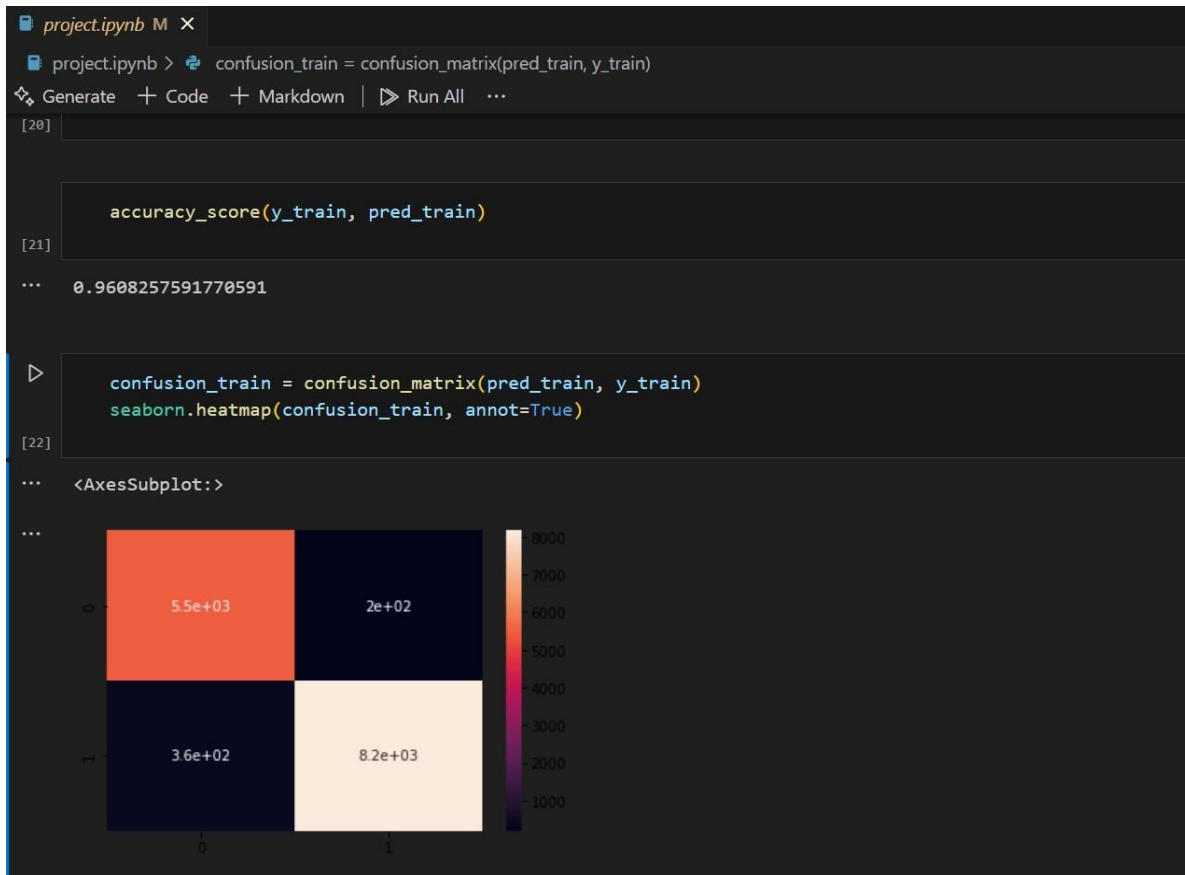
- We used the Fake News dataset, which contains 21,000+ labeled news articles. Each record includes fields such as:
- Title: The headline of the news article.
- Text: The main content of the article.
- Subject: The fake news detection

Label: 1 for fake news and 0 for real news.

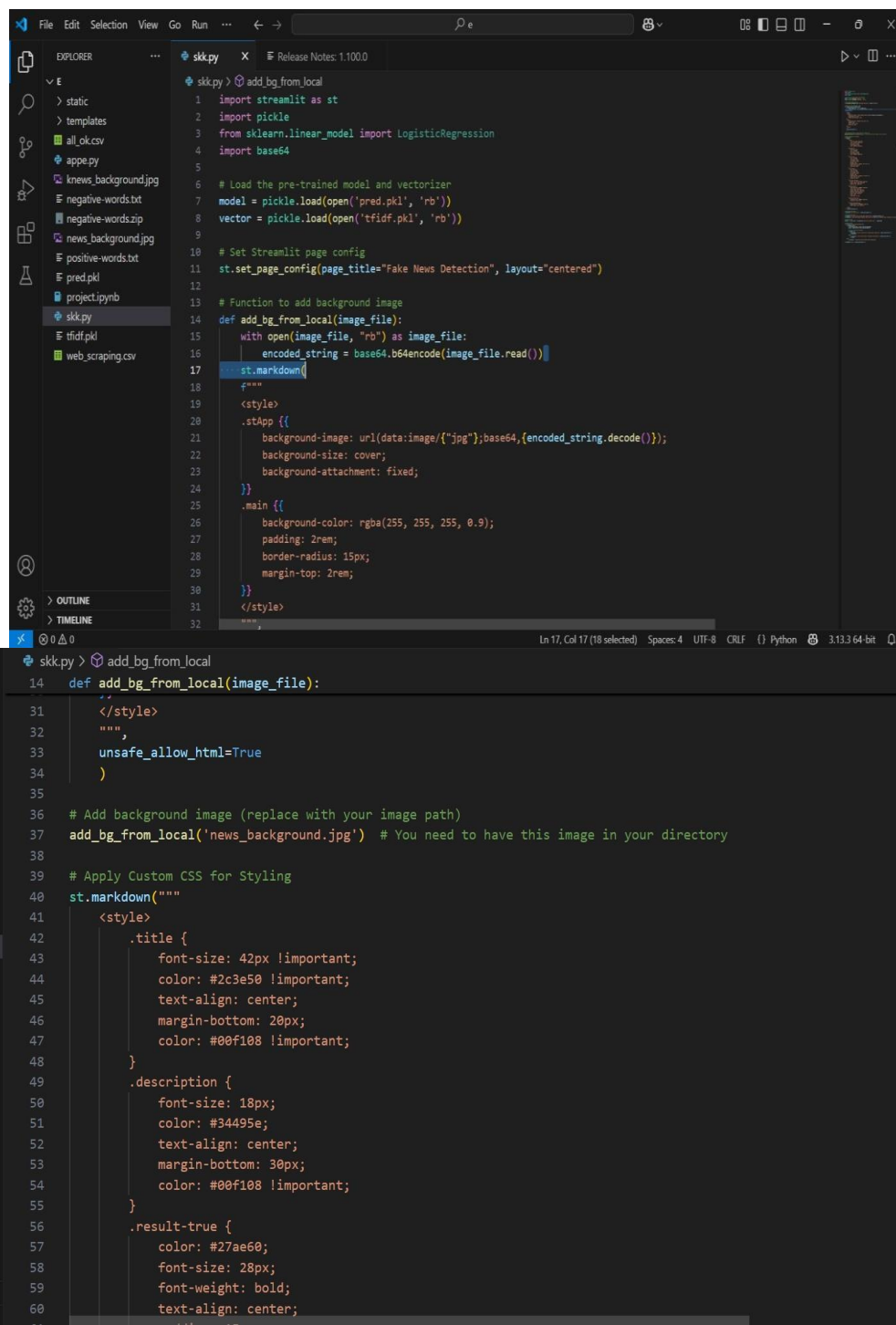
- Other potential datasets such as the LIAR dataset provide additional granularity, including metadata about the source, author, and context, which can enhance model robustness.
- Data Preprocessing
- Preprocessing is crucial for converting raw text into a format suitable for machine learning. The following steps were applied:
- Lowercasing: Convert all text to lowercase.
- Punctuation Removal: Remove all punctuation marks.
- Stop Words Removal: Filter out common words (e.g., 'the', 'is', 'and') that do not contribute to meaning.
- Tokenization: Split text into individual words or tokens.
- Lemmatization: Convert words to their base or dictionary form.
- Noise Removal: Eliminate URLs, HTML tags, and special characters.

- In addition, we explored the use of advanced techniques such as named entity recognition (NER) and part-of-speech (POS) tagging to understand syntactic and semantic elements of the text.
- Feature Extraction
- To convert text into numerical features, we used:
- Bag of Words (BoW): Represents text by word frequency.
- TF-IDF (Term Frequency-Inverse Document Frequency): Highlights important words in the corpus.
- **Word Embeddings** : Captures semantic meaning using Word2Vec and GloVe. Pre-trained embeddings help in generalizing better across datasets.
- Feature selection methods such as chi-square and mutual information were used to identify the most informative features, reducing dimensionality and improving model performance.
- Model Building
- We implemented and tested the following models:
- Logistic Regression: A simple linear model for binary classification.
- Random Forest: An ensemble model that combines decision trees for robust predictions.
- **Support Vector Machine (SVM)** : Effective in high-dimensional text classification.
- **LSTM** : A deep learning model that handles sequential dependencies in text.
- **BERT** : A transformer-based model pre-trained on large corpora. Fine-tuned on our dataset for high accuracy.
- Hyperparameter tuning was performed using grid search and cross-validation techniques to identify the optimal parameters for each model.
- Model Training and Evaluation
- The dataset was split into training (67%) and testing (33%) sets. Stratified sampling ensured balanced class distribution. We employed k-fold cross-validation to further validate the robustness of our models.
- Evaluation metrics included:
- **Accuracy**: Overall correctness of the model.
- **Precision**: Correctness of positive predictions.
- **Recall** : Coverage of actual positive cases.
- **F1-Score** : Harmonic mean of precision and recall.
- **Confusion Matrix** : Detailed breakdown of true/false positives and negatives.

- **ROC-AUC Score** : Measures the ability of the classifier to distinguish between classes.



Complete Code:-



```
File Edit Selection View Go Run ... < -> e
EXPLORER
  E
    > static
    > templates
    all_ok.csv
    app.py
    news_background.jpg
    negative-words.txt
    negative-words.zip
    news_background.jpg
    positive-words.txt
    pred.pkl
    project.ipynb
    skk.py
    tfidf.pkl
    web_scraping.csv
  OUTLINE
  TIMELINE
  0 0 0
  Ln 17, Col 17 (18 selected) Spaces: 4 UTF-8 CRLF Python 3.13.3 64-bit

skk.py > add_bg_from_local
1 import streamlit as st
2 import pickle
3 from sklearn.linear_model import LogisticRegression
4 import base64
5
6 # Load the pre-trained model and vectorizer
7 model = pickle.load(open('pred.pkl', 'rb'))
8 vector = pickle.load(open('tfidf.pkl', 'rb'))
9
10 # Set Streamlit page config
11 st.set_page_config(page_title="Fake News Detection", layout="centered")
12
13 # Function to add background image
14 def add_bg_from_local(image_file):
15     with open(image_file, "rb") as image_file:
16         encoded_string = base64.b64encode(image_file.read())
17     st.markdown(
18         """
19         <style>
20         .stApp {
21             background-image: url(data:image/{format};base64,{encoded_string.decode()});
22             background-size: cover;
23             background-attachment: fixed;
24         }
25         .main {
26             background-color: rgba(255, 255, 255, 0.9);
27             padding: 2rem;
28             border-radius: 15px;
29             margin-top: 2rem;
30         }
31         </style>
32         """,
33         unsafe_allow_html=True
34     )
35
36 # Add background image (replace with your image path)
37 add_bg_from_local('news_background.jpg') # You need to have this image in your directory
38
39 # Apply Custom CSS for Styling
40 st.markdown("""
41     <style>
42     .title {
43         font-size: 42px !important;
44         color: #2c3e50 !important;
45         text-align: center;
46         margin-bottom: 20px;
47         color: #00f108 !important;
48     }
49     .description {
50         font-size: 18px;
51         color: #34495e;
52         text-align: center;
53         margin-bottom: 30px;
54         color: #00f108 !important;
55     }
56     .result-true {
57         color: #27ae60;
58         font-size: 28px;
59         font-weight: bold;
60         text-align: center;
61     }
62     """,
63     unsafe_allow_html=True
64 )
```

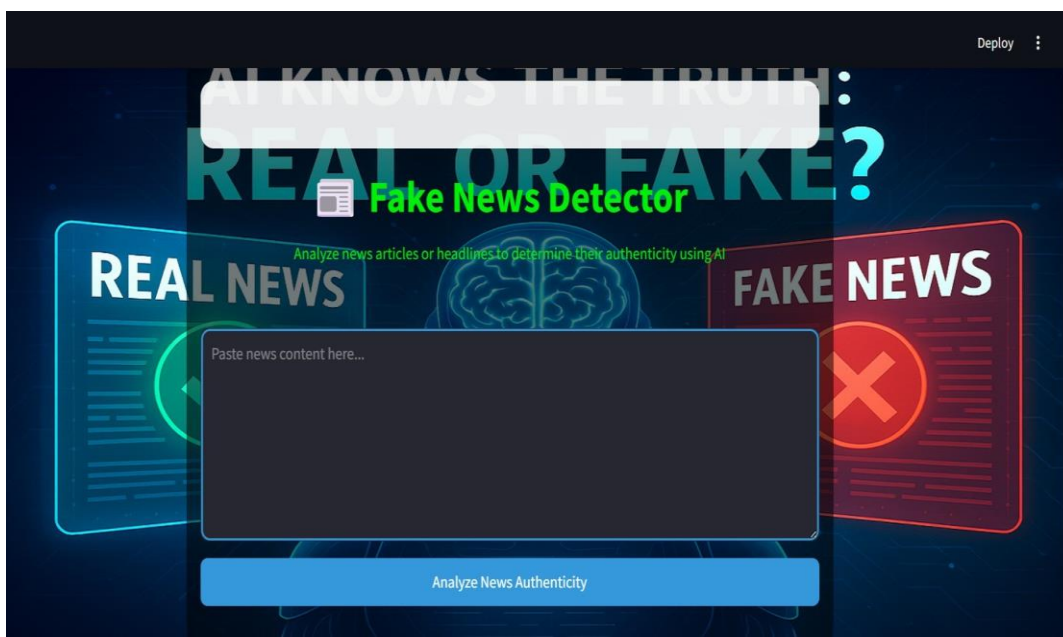
```
Go Run ... < -> e
skk.py x Release Notes: 1.100.0
skk.py > add_bg_from_local
59 font-weight: bold;
60 text-align: center;
61 padding: 15px;
62 background-color: rgba(39, 174, 96, 0.1);
63 border-radius: 10px;
64 margin-top: 20px;
65 }
66 .result-fake {
67 color: #e74c3c;
68 font-size: 28px;
69 font-weight: bold;
70 text-align: center;
71 padding: 15px;
72 background-color: rgba(231, 76, 60, 0.1);
73 border-radius: 10px;
74 margin-top: 20px;
75 }
76 .stTextArea textarea {
77 border: 2px solid #3498db !important;
78 border-radius: 10px !important;
79 padding: 10px !important;
80 }
81 .stButton>button {
82 background-color: #3498db !important;
83 color: white !important;
84 border-radius: 8px !important;
85 padding: 10px 24px !important;
86 font-size: 18px !important;
87 border: none !important;
88 width: 100%;
89 transition: all 0.3s;
90 }
```

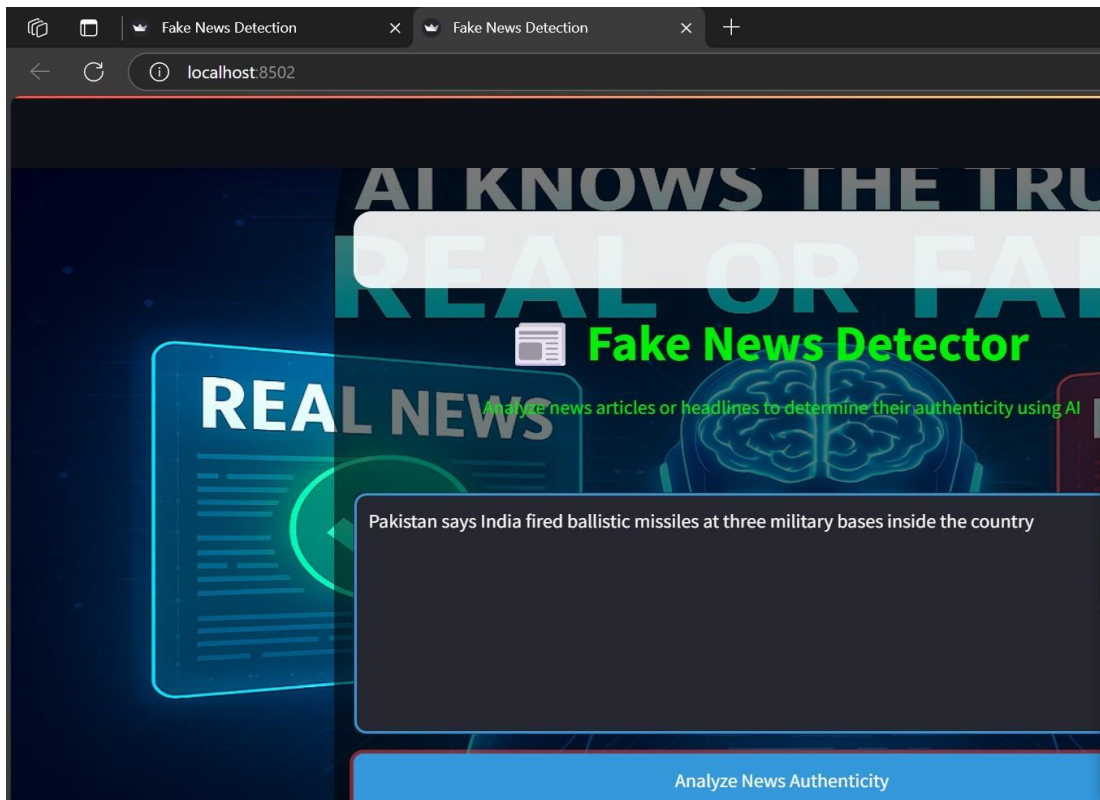
Ln 17, Col 17 (18 selected) Spaces: 4 UTF-8 CRLF () Python 3.1

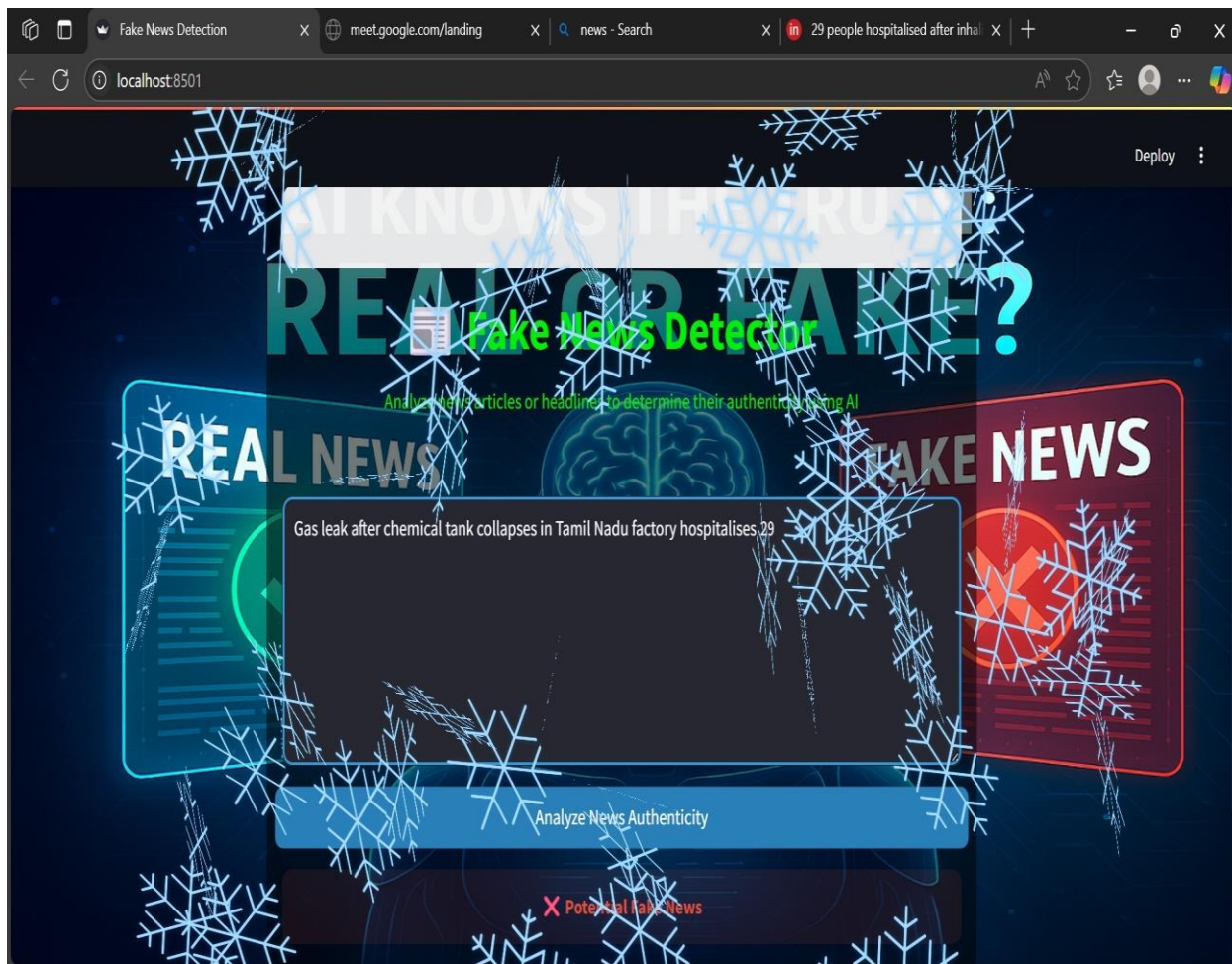
```
ew Go Run ... < -> e
... skk.py x Release Notes: 1.100.0
skk.py > add_bg_from_local
90 }
91 .stButton>button:hover {
92 background-color: #2980b9 !important;
93 transform: scale(1.02);
94 }
95 .st-emotion-cache-mtjnbi {
96 background-color: #00000087 !important;
97 border-radius: 20% !important;
98 }
99 </style>
100 "", unsafe_allow_html=True)
101
102 # Main content container
103 st.markdown("<div class='main'>", unsafe_allow_html=True)
104
105 # Title and Description
106 st.markdown("<h1 class='title'>🚫 Fake News Detector</h1>", unsafe_allow_html=True)
107 st.markdown("<p class='description'>Analyze news articles or headlines to determine their authenticity using AI</p>")
108
109 # Input text area
110 message = st.text_area("", placeholder="Paste news content here...", height=200)
111
112 # Predict button
113 if st.button("Analyze News Authenticity"):
114     if message.strip():
115         transformed_message = vector.transform([message])
116         output = model.predict(transformed_message)[0]
117
118         # Display result with animation
119         if output == 1:
120             st.markdown("<p class='result-true'>✅ Genuine News Content</p>", unsafe_allow_html=True)
121             st.balloons()
```

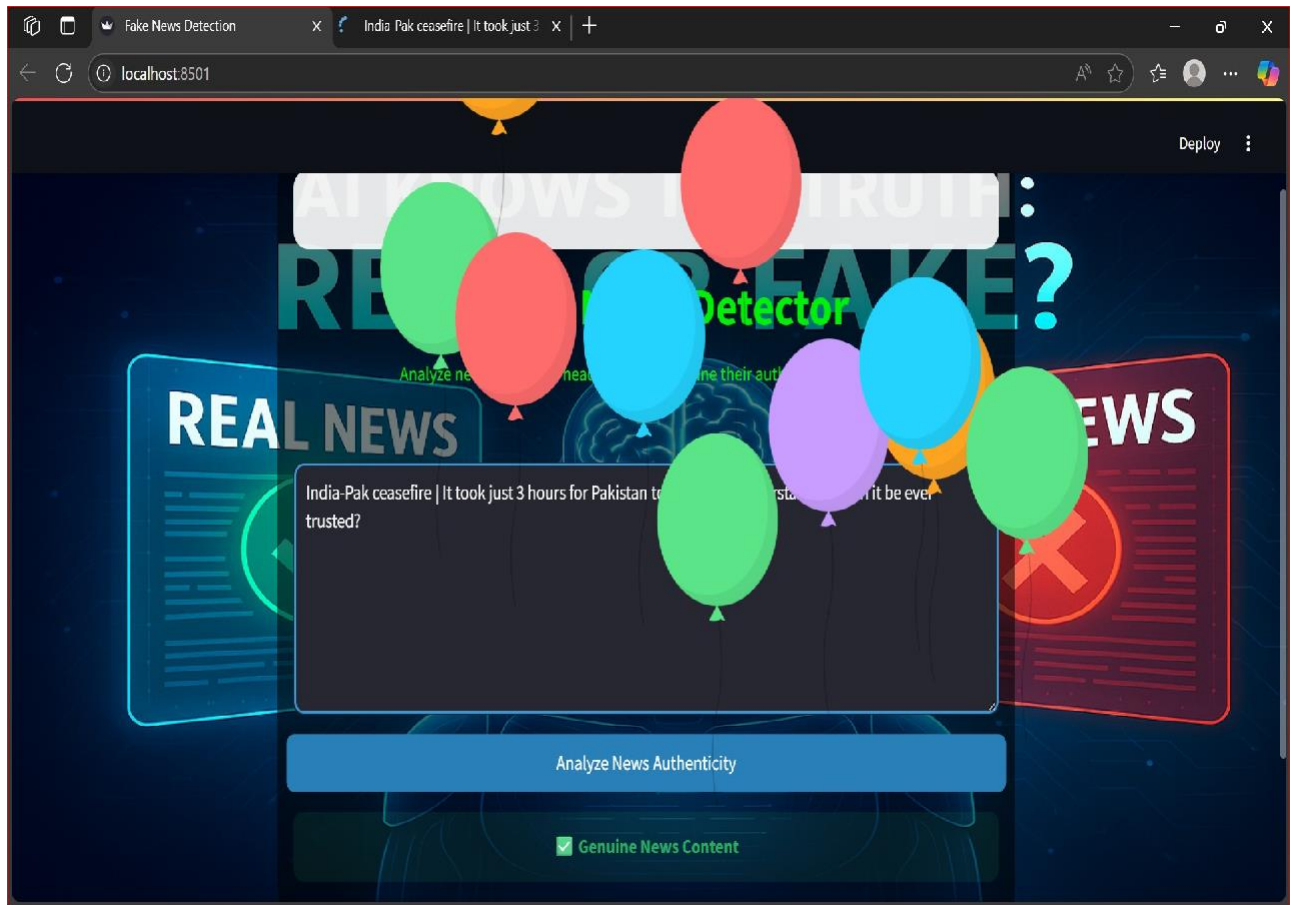
```
skk.py x Release Notes: 1.100.0
skk.py > add_bg_from_local
111
112 # Predict button
113 if st.button("Analyze News Authenticity"):
114     if message.strip():
115         transformed_message = vector.transform([message])
116         output = model.predict(transformed_message)[0]
117
118         # Display result with animation
119         if output == 1:
120             st.markdown("<p class='result-true'>✅ Genuine News Content</p>", unsafe_allow_html=True)
121             st.balloons()
122         else:
123             st.markdown("<p class='result-fake'>❌ Potential Fake News</p>", unsafe_allow_html=True)
124             st.snow()
125     else:
126         st.warning("⚠ Please enter news content before analyzing")
127
128 st.markdown("</div>", unsafe_allow_html=True)
```

Output Cheak:-









CONCLUSION:

Through this project, I have tried to provide brief information on how Emotion Recognition works using Deep Learning. We trained our Convolutional Neural Network Model on top of pre-

trained data (ResNet50) and we also used Computer Vision as part of this model. Haarcascade is the package used from OpenCV to detect objects in other images.

We trained the model with several images and then used the test images to see how the results match up. We trained the model through epochs. In this model, we have taken epochs as 100. Once the threshold is achieved by the model and if we further try to train our model, then it will provide unexpected results and its accuracy will also decrease. After that, increasing the epoch would also not help. Hence, epochs play a very important role in deciding the accuracy of the model, and its value can be decided through trial and error.

For this model, the accuracy that we achieved for the validation set is 63%. To further increase the accuracy of the model, we can either expand the training dataset we have or increase the step size for the model. Through these parameters, we can increase the model accuracy for this model.